Benoît Bonnet-Weill

Chargé de Recherche CNRS

Junior CNRS Researcher

(Last updated on December 22, 2025)

Personal information

Civil Status: Born on the 27th of April 1993 in Paris, XII$^{\text{ième}}$ arrondissement.
Happily married since the 25th of August 2018.
Father of two lovely daughters born in July 2023 and July 2025.

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Google Scholar: https://scholar.google.fr/citations?user=Ow5eQawAAAAJ&hl=fr

Academic positions

- January 2024 – Now: **Junior CNRS Researcher** in the team Modélisation, Estimation et Analyse des Systèmes (MODESTY), *Laboratoire des Signaux et Systèmes*, Gif-sur-Yvette.
- November 2021 – December 2023: **Junior CNRS Researcher** in the team Méthodes et Algorithmes de Commande (MAC), *Laboratoire d'Analyse et d'Architecture des Systèmes*, Toulouse.
- February 2021 – October 2021: **INRIA Postdoctoral Fellow** under the supervision of Mario Sigalotti and Nastassia Pouradier Duteil, *Laboratoire Jacques-Louis Lions*, Paris.
- November 2019 – February 2021: **CNRS Postdoctoral Fellow** under the supervision of Hélène Frankowska, *Institut de Mathématiques de Jussieu - Paris Rive Gauche*, Paris.
- October 2016 – October 2019: **Ph.D. Student in Applied Mathematics** under the supervision of Francesco Rossi and Maxime Hauray, *Laboratoire d'Informatique et Systèmes*, Marseille & *Università degli Studi di Padova*, Padova.

Education

- October 2016 – October 2019: **Ph.D. in Applied Mathematics**. Specialisation in Control Theory.
  - **Title:** Optimal Control in Wasserstein Spaces
  - **Advisers:** Francesco Rossi (Director), *Università degli Studi di Padova*, Padova.
    Maxime Hauray (Codirector), *Aix-Marseille Université*, Marseille.
  - **Jury:** Filippo Santambrogio (President), *Université Claude Bernard*, Lyon.
    Pierre Cardaliaguet (Referee), *Université Paris-Dauphine*, Paris.
    Nicola Gigli (Referee), *Scuola Internazionale Superiore Degli Studi Avanzati*, Trieste.
    José Antonio Carrillo (Examinator), *Oxford University*, Oxford.
    Hélène Frankowska (Examinator), *CNRS* & *Institut de Mathématiques de Jussieu*, Paris.

Francesca Chittaro (Examinator), *Laboratoire d'Informatique et Systèmes*, Toulon.
Francesco Rossi (Director), *Università degli Studi di Padova*, Padova.
Maxime Hauray (Codirector), *Aix-Marseille Université*, Marseille.
Jean-Paul Gauthier (Invited), *Laboratoire d'Informatique et Systèmes*, Toulon.

- September 2015 – August 2016: **M.Sc. in Applied Mathematics**. Specialisation in Optimisation, Calculus of Variations and Geometric Control. *Université Paris-Saclay*, Orsay.
- September 2013 – September 2016: **Engineering curriculum in Applied Mathematics**. Specialisation in Optimisation, Control Theory and Operational Research. *École Nationale Supérieure de Techniques Avancées (ENSTA Paris)*, Palaiseau.
- September 2011 – September 2013: **"Classes Préparatoires aux Grandes Écoles"** with Mathematics and Physics majors (MPSI - MP*). *Lycée Blaise Pascal*, Orsay.
- September 2008 – September 2011: **High School Education** with Mathematics and Physics majors ($1^{\text{ère}}$ - Terminale S). *Lycée Descartes*, Antony.

Community services

- January 2024 – Now: **Coorganiser** of the *Séminaire d'Automatique du Plateau de Saclay*.
- July 2023 – Now: **Elected member of the *Conseil Scientifique d'Institut* (CSI)** of the CNRS National Institute of Mathematics and its Interactions (INSMI) – Electoral College B1.

Grants

- February 2024 – December 2024: **Functional support** of 3000€ granted by the H-Code interdisciplinary object operating at the level of Université Paris-Saclay.
- February 2021 – October 2021: **15-month Postdoctoral Fellowship** from INRIA (interrupted to take my position at CNRS), *Université Pierre et Marie Curie*, Paris.
- October 2016 – October 2019: **3-year Ph.D. funding** from the Archimède French Excellence Laboratory, *Laboratoire d'Informatique et Systèmes*, Marseille.

Conference organisations

[O2] CIRM Workshop *Variational Analysis, Models and Methods in Measure Spaces – Control, Optimisation and Learning in Large-Scale Systems* (with Massimo Fornasier, Hélène Frankowska and Giuseppe Savaré) – Centre International de Rencontres Mathématiques (CIRM), Marseille (April 29th - May 3rd 2024).

[O1] Mini-course *Measure differential equations: modelling and numerical solutions* (with Didier Henrion, Swann Marx and Francesco Rossi) – 22nd Symposium on Mathematical Theory of Networks and Systems (MTNS2022), Bayreuth (September 14th 2022).

Publications

The preprints of my articles can be found on my Homepage or via my Google Scholar account.

Submitted and under-revision

[S3] R. Bonalli, B. Bonnet-Weill and L. Pfeiffer. A Characterization of Law-Invariant and Coherent Risk Measures through Optimal Transport. *Submitted*, 2025.

[S2] B. Bonnet-Weill and N. Pouradier Duteil. Nonlocal Continuity Equations in Fibred Wasserstein Spaces. *Submitted*, 2025.

[S1] B. Bonnet-Weill, A. Corella and H. Frankowska. Viability Theory in the 1-Wasserstein Space. *Submitted*, 2025.

Published and accepted journal papers

[J14] B. Bonnet-Weill and M. Korda. Set-Valued Koopman Theory for Control Systems. *SIAM Journal on Control and Optimization*, 63(5):3644-3673, 2025.

[J13] B. Bonnet-Weill and H. Frankowska. Carathéodory Theory and A Priori Estimates for Continuity Inclusions in the Space of Probability Measures. *Nonlinear Analysis, Theory, Methods and Applications*, 247:113595, 2024.

[J12] B. Bonnet-Weill and H. Frankowska. On the Viability and Invariance of Proper Sets for Continuity Inclusions in Wasserstein Spaces. *SIAM Journal on Mathematical Analysis*, 56(3):2863-2914, 2024.

[J11] R. Bonalli and B. Bonnet. First-Order Pontryagin Optimality Conditions for Risk-Averse Stochastic Optimal Control Problems. *SIAM Journal on Control and Optimization*, 61(3):1881-1909, 2023.

[J10] B. Bonnet, C. Cipriani, M. Fornasier and H. Huang. A Measure Theoretical Approach to the Mean-Field Maximum Principle for Training NeurODEs. *Nonlinear Analysis*, 227:113161, 2023.

[J9] B. Bonnet, N. Pouradier Duteil and M. Sigalotti. Consensus Formation in First-Order Graphon Models with Time-Varying Topologies. *Mathematical Models and Methods in Applied Sciences*, 32(11):2121-2188, 2022.

[J8] B. Bonnet and H. Frankowska. Semiconcavity and Sensitivity Analysis in Mean-Field Optimal Control and Applications. *Journal de Mathématiques Pures et Appliquées*, 157:282-345, 2022.

[J7] B. Bonnet and H. Frankowska, Necessary Optimality Conditions for Optimal Control Problems in Wasserstein Spaces. *Applied Mathematics and Optimization*, 84:1281-1330, 2021.

[J6] B. Bonnet and F. Rossi. Intrinsic Lipschitz Regularity of Mean-Field Optimal Controls. *SIAM Journal on Control and Optimization*, 59(3):2011-2046, 2021.

[J5] B. Bonnet and É. Flayac, Consensus and Flocking Under Communication Failures for a Class of Cucker-Smale Systems. *Systems and Control Letters*, 152:104930, 2021.

[J4] B. Bonnet and H. Frankowska. Differential Inclusions in Wasserstein Spaces: The Cauchy-Lipschitz Framework. *Journal of Differential Equations*, 271:594-637, 2021.

[J3] B. Bonnet. A Pontryagin Maximum Principle in Wasserstein Spaces for Constrained Optimal Control Problems. *ESAIM COCV*, 25(52), 2019.

[J2] B. Bonnet, J.P. Gauthier and F. Rossi. Generic Singularities of the 3D-Contact Conjugate Locus. *Comptes Rendus Mathématiques*, 357(6):520-527, 2019.

[J1] B. Bonnet and F. Rossi. The Pontryagin Maximum Principle in the Wasserstein Space. *Calculus of Variations and Partial Differential Equations* 58:11, 2019.

Conference proceedings

[C6] B. Bonnet-Weill and M. Sigalotti. Exponential Consensus Formation in Time-Varying Multiagent Systems via Compactification Methods. *2024 63rd IEEE Conference on Decision and Control (CDC), 5409-5416*, 2024.

[C5] B. Bonnet and H. Frankowska. Viability and Exponentially Stable Trajectories for Differential Inclusions in Wasserstein Spaces. *2022 61st IEEE Conference on Decision and Control (CDC), 5086-5091*, 2022.

[C4] B. Bonnet and H. Frankowska. On the Properties of the Value Function Associated to a Mean-Field Optimal Control Problem of Bolza Type. *2021 IEEE Conference on Decision and Control (CDC)*, 4558-4563, 2021.

[C3] B. Bonnet and F. Rossi. Variance Optimization and Control Regularity for Mean-Field Dynamics. *IFAC-PapersOnLine*, 54 (19):13-18, 2021.

[C2] B. Bonnet and H. Frankowska. Mean-Field Optimal Control of Continuity Equations and Differential Inclusions. *2020 IEEE Conference on Decision and Control (CDC)*, 470-475, 2020.

[C1] B. Bonnet and F. Rossi. Sparse Control of Kinetic Cooperative Systems to Approximate Alignment. *Proceedings of the 20th IFAC World Congress*, 2017.

Lecture notes

[L1] B. Bonnet-Weill. Théorie de la Mesure et Intégration de Lebesgue – Approches Géométriques et Fonctionnelles. *Module Thématique AOT11*, ENSTA Paris, 2025.

Presentations at conferences and seminars

Invited talks at conferences, seminars and workshops

[I25] *Structured Continuity Equations in Fibered Probability Spaces* – Round MeanField IV: N-Body sul Canal Grande, Venezia (September 2025).

[I24] *Structured Continuity Equations in Fibered Wasserstein Spaces* – Rutgers Camden Math Seminar Series, Philadelphia (March 2025).

[I23] *Consensus Formation in Graphon Dynamics: Motsch and Tadmor Revisited* – Workshop in the honour of E. Tadmor on Emergent Behavior in Complex Systems of Interacting Agents, Institute for Mathematics and Statistics Innovation, Chicago (March 2025).

[I22] *A Set-Valued Approach to Koopman Operators for Control Systems* – Department of Mathematics and Center for Complex Systems, University of Namur (March 2024).

[I21] *Measure Dynamics and Macroscopic Approximations of Multi-Agent Systems* – Fédération de Mathématiques CentraleSupélec, Plateau de Saclay (December 2023).

[I20] *Set-Valued Dynamics with State-Constraints in Wasserstein Spaces* – Workshop on Optimal Transport, Mean-Field Models, and Machine Learning, Technische Universtität München (April 2023).

[I19] *Set-Valued Generalisations of Koopman Operators* – Workshop on Real Algebraic Geometry with a View Towards Koopman Operator Methods, Oberwolfach (March 2023).

[I18] *New Results on Asymptotic Consensus Formation in Graphon Dynamics* – Séminaire d'Automatique du Plateau de Saclay, Laboratoire des Signaux et Systèmes, Centrale Supélec, Gif-sur-Yvette (February 2023).

[I17] *On the Lipschitz Regularity of Mean-Field Optimal Controls* – Groupe de Travail Contrôle, Laboratoire Jacques-Louis Lions, Sorbonne-Université, Paris (January 2023).

[I16] *Pontryagin Optimality Conditions in Wasserstein Spaces and their Application to the Training of NeurODEs* – Séminaire d'Analyse Non Linéaire et d'Optimisation, Avignon Université, Avignon (October 2022).

[I15] *Some Results Related to Consensus Formation in Graphon Dynamics* – Conference "Round Mean-field: Crowds, Opinions, Cells", LYSM, Roma (September 2022).

[I14] *When HJB Meets Pontryagin in Mean-Field Control* – Invited session "Optimal Control and Calculus of Variations on Metric Spaces", 15th Viennese Conference on Optimal Control and Dynamics Games, Vienna (July 2022)

[I13] *A Mean-Field Optimal Control Approach to Deep Learning* – Invited session "Contrôle et Jeux à Champ-Moyen", Journées SMAI MODE, Limoges (June 2022).

[I12] *Consensus Formation, Macroscopic Approximations, and their Interactions in the context of Multi-Agent Dynamics* – DO Seminar, LAAS-CNRS, Toulouse (May 2022).

[I11] *Set-Valued Dynamics in the Space of Probability Measures* – Journée Rencontre de l'Équipe Combinatoire et Optimisation, IMJ-PRG, Paris (April 2022).

[I10] *A Mean-Field Optimal Control Approach to the Training of NeurODEs* – BrainPOP Seminar, LAAS-CNRS, Toulouse (January 2022).

[I9] *Fine Properties of the Value Function in Mean-Field Optimal Control* – Invited session "Mean-Field Games and Applications", PGMO Days, Palaiseau (December 2021).

[I8] *Nonsmooth and Set-Valued Analysis in Wasserstein Spaces with Applications in Mean-Field Control* – Séminaire Parisien d'Optimisation, IHP, Paris (November 2021).

[I7] *Sufficient Conditions for the Lipschitz Regularity of Mean-Field Optimal Controls* – Groupe de Travail de Calcul des Variations, Remote talk (March 2021).

[I6] *Exponential Flocking under Communication Failures for some Cucker-Smale Models* – Seminar of the INRIA Team MAMBA, LJLL, Remote talk (March 2021).

[I5] *Continuity Inclusions and Applications in Mean-Field Optimal Control* – Seminar of Analysis and Applications, Université de Bretagne Occidentale, Remote talk (February 2021).

[I4] *Flocking for the Cucker-Smale Systems under Communication Failures* – Seminar of INRIA Team CAGE, LJLL, Remote talk (May 2020).

[I3] *Intrinsic Lipschitz Regularity in Mean-Field Optimal Control Problems* – Seminar of Probability, Statistics and Control Theory, ENSTA Paris, Palaiseau (October 2019).

[I2] *Topics in Analysis and Optimal Control of Multi-Agent Systems* – Seminario di Equazioni Differenziale, Università degli Studi di Padova, Padova (March 2019).

[I1] *Optimal Control Problems in Wasserstein Spaces* – Invited session "Variational Analysis and Optimal Control", 14th Viennese Conference on Optimal Control and Dynamics Games, Vienna (August 2018).

Presentations at international conferences and research schools

[P10] *Exponential Consensus Formation via Compactification Methods* – 63rd IEEE Conference on Decision and Control, Milano (December 2024).

[P9] *A Meanfield Control Perspective on the Training of ResNets* – French-German-Spanish Conference on Optimization, Gijon (June 2024).

[P8] *Set-Valued Koopman Theory for Control Systems* – 2022 International Symposium on Nonlinear Theory and its Applications, Remote talk (December 2022).

[P7] *Macroscopic Approximations of Multi-Agent Systems: An Introduction to Continuity Equations* – Mini-course on Measure Differential Equations, 25th International Symposium on Mathematical Theory of Networks and Systems, Bayreuth (September 2022).

[P6] *Variance Optimization and Control Regularity in Mean-Fields Dynamics* – 7th IFAC Workshop on Lagrangian and Hamiltonian Methods for Nonlinear Control, Remote talk (October 2021).

[P5] *Mean-Field Control and Continuity Inclusions* – 59th IEEE Conference on Decision and Control, Remote talk (December 2020).

[P4] *Some Problems in Modelling and Optimal Control of Multi-Agent Systems* – Poster session at the conference Crowds: models and control, CIRM, Marseille (June 2019).

[P3] *Optimal Control of Multi-Agent Systems: A Pontryagin Approach* – Toulouse Winter School in Calculus of Variations and Probability Theory, IMT, Toulouse (February 2019).

[P2] *Optimal Control Problems in Wasserstein Spaces* – 12th International Young Researcher Workshop on Geometry, Mechanics and Control, Università degli Studi di Padova, Padova (January 2018).

[P1] *Sparse Alignment of Kinetic Cooperative Systems* – 20th IFAC World Congress, Toulouse (July 2017).

Editorial activities

Reviewer for the journals *Probability Theory and Related Fields*, *SIAM Journal on Control and Optimization*, *SIAM Journal on Mathematics of Data Sciences*, *Journal of Differential Equations*, *Mathematics of Computations*, *ESAIM COCV*, *Applied Mathematics and Optimization*, *Automatica*, *IEEE Transactions on Automatic Control*, *Systems and Control Letters*, *Journal of Mathematical Analysis and Applications*, *Journal of Dynamical and Control Systems*, as well as for the proceedings of the *IEEE Conference on Decision and Control*, *American Control Conference* and *IFAC World Congress*.

Teaching activities

- 2023 – Now: **Lectures** for the course Measure Theory and Lebesgue Integration.
  Bachelor 3 level (specialisation), 27-hour teaching, *ENSTA Paris*, Palaiseau.
- 2019 – 2023: **Exercises sessions** for the course Differentiable Optimisation I.
  Master 1 level, 15-hour teaching, *ENSTA Paris* & *UPSAY*, Palaiseau.
- 2020 – 2021: **Exercises sessions** for the course Differentiable Optimisation II.
  Master 1 level, 15-hour teaching, *ENSTA Paris* & *UPSAY*, Palaiseau.
- 2020 – 2021: **Exercises sessions** for the course Optimisation.
  Bachelor 3 level, 18-hour teaching, *UPP1*, Paris.

- 2019 – 2020: **Exercises sessions** for the course Quadratic Optimisation.
  Bachelor 3 level, 15-hour teaching, *ENSTA Paris*, Palaiseau.
- 2017 – 2019: **Lectures** for the course Introduction to Lebesgue Integration.
  Bachelor 3 level, 4-hour teaching, *ECM*, Marseille.
- 2017 – 2019: **Lectures** for the course Introduction to Optimisation Theory.
  Bachelor 3 level, 2-hour teaching, *ECM*, Marseille.
- 2017 – 2018: **Exercises sessions** for the course Preliminaries and Recalls in Optimisation.
  Master 2 level, 14-hour teaching, *ECM* & *AMU*, Marseille.

Miscellaneous Skills, Hobbies and Interests

- **Languages**
  - French (Mother tongue)
  - Italian (Basic, lived in Italy for a while)
  - English (Fluent, C2-level CEFR)
  - German & Chinese (Small remnants)
- **Hobbies**
  - Drums (15-year regular practice)
  - Billiard (6-year regular practice)
  - Chess (2-year somewhat practice)
  - Bodhran (celtic traditional drums) & Guitar
  - Bouldering (indoor climbing) & bike travels
  - Video games, board games & card games
- **Interests**
  - Epistemology of mathematics & physics
  - Sociology & political philosophy
  - Science fiction & fantasy novels
  - Climate sciences (MyCO2 ambassador)
  - Music in general
  - “Bandes dessinées” and mangas